F1 Insight API

A Comprehensive Formula 1 Historical Data Analysis API

Covering Every Season from 1950 to 2024

Module: **COMP3011 Web Services and Web Data**

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Technology Stack: **FastAPI · SQLAlchemy · PostgreSQL · JWT Auth**

Deployment: **Render (Cloud) + Neon PostgreSQL**

Live API: https://f1-insight-api.onrender.com

18	**10**	**26,000+**	**850+**
Total Endpoints	Data Tables	Race Results	Drivers

1. API Architecture Overview

The F1 Insight API is a RESTful service built with **FastAPI** (Python) following a layered architecture: Router → Service → Repository → Database. The API adheres to OpenAPI 3.1 specification with full auto-generated interactive documentation via Swagger UI.

Architecture Highlights

- Layered design with clear separation of concerns (routers, services, models, schemas)
- SQLAlchemy ORM with async-compatible session management
- Pydantic v2 schemas with strict validation, examples, and field descriptions
- JWT Bearer authentication via OAuth2 password flow
- Role-based access control (user / admin) for write and delete operations
- Cloud-native deployment on Render with Neon serverless PostgreSQL

2. Endpoint Summary

The API exposes 18 endpoints across 5 functional groups, covering CRUD operations, analytics, and advanced predictive models.

2.1 Authentication

Secure user registration and login flow issuing JWT tokens for subsequent authenticated requests.

Method	Endpoint	Description
POST	/api/v1/auth/register	Register a new user account
POST	/api/v1/auth/login	Login and receive JWT Bearer token

2.2 Drivers CRUD

Full lifecycle management for 850+ F1 drivers with pagination, multi-field search, nationality filtering, and 5 smart sort modes.

Method	Endpoint	Description
GET	/api/v1/drivers/	Paginated list with filters and sorting
POST	/api/v1/drivers/	Create driver (auth required)
GET	/api/v1/drivers/{driver_id}	Get driver by ID
PUT	/api/v1/drivers/{driver_id}	Update driver (auth required)
DELETE	/api/v1/drivers/{driver_id}	Delete driver (admin only)

2.3 Circuits CRUD

Circuit management with geolocation data (lat/lng/alt), country filtering, and detail views showing the 5 most recent race winners.

Method	Endpoint	Description
GET	/api/v1/circuits/	Paginated list with stats
POST	/api/v1/circuits/	Create circuit (auth required)
GET	/api/v1/circuits/{circuit_id}	Detail with recent race history
PUT	/api/v1/circuits/{circuit_id}	Update circuit (auth required)
DELETE	/api/v1/circuits/{circuit_id}	Delete circuit (admin only)

2.4 Constructors CRUD

Constructor (team) management with aggregated career stats and top-5 driver breakdowns in detail views.

Method	Endpoint	Description
GET	/api/v1/constructors/	Paginated list with points and wins
POST	/api/v1/constructors/	Create constructor (auth required)
GET	/api/v1/constructors/{id}	Detail with career summary and top drivers
PUT	/api/v1/constructors/{id}	Update constructor (auth required)
DELETE	/api/v1/constructors/{id}	Delete constructor (admin only)

2.5 Analytics Endpoints

Core analytical capabilities computed in real time from raw race data — no pre-aggregated tables.

Method	Endpoint	Description
GET	`.../drivers/{id}/career-stats`	Lifetime stats: races, wins, podiums, poles, DNFs, rates
GET	`.../drivers/{id}/season-progression`	Round-by-round cumulative points curve
GET	`.../drivers/compare?d1=&d2=`	Head-to-head comparison with direct encounters
GET	`.../races/{id}/pit-stop-analysis`	Per-driver stop count, pit time, strategy summary
GET	`.../circuits/{id}/history`	Total races, year range, top 5 winningest drivers

2.6 Advanced Analytics

Sophisticated models that go beyond simple aggregation, providing predictive and multi-dimensional insights.

Method	Endpoint	Description
GET	`.../win-probability?driver_id=&circuit_id=`	4-factor weighted win probability prediction
GET	`.../drivers/{id}/performance-summary`	5-dimension performance rating (0–100)
GET	`.../teammate-battle?driver_id=&season=`	Qualifying + race + points vs teammate(s)
GET	`.../leaderboard?metric=&season=`	Dynamic ranking by points/wins/podiums/win_rate

3. Key Design Highlights

3.1 Win Probability Prediction Model

A custom 4-factor weighted model that predicts a driver's probability of winning at a specific circuit, returning a 0–100% score with full factor breakdown:

Factor	Weight	What It Measures
Circuit History	35%	Win rate, podium rate, and average finish at this specific circuit
Recent Form	30%	Last 5 races: average finish, wins, podiums
Grid Conversion	20%	Front-row starts converted to victories
Career Baseline	15%	Overall career win rate as a floor estimate

3.2 Multi-Dimensional Performance Rating

Each driver receives a composite score (0–100) based on five weighted dimensions, enabling fair comparisons across eras:

Dimension	Weight	Metrics Used
Qualifying Pace	25%	Avg qualifying position, poles, top-3 qualifying rate
Race Pace	30%	Avg finish position, wins, podiums, points per race
Consistency	20%	Finish rate, points-scoring rate, DNF count
Overtaking Ability	15%	Average positions gained (grid → finish)
Pit Stop Efficiency	10%	Median pit stop time across career

3.3 Smart Sorting and Filtering

All three CRUD resources (Drivers, Circuits, Constructors) support rich query capabilities that go beyond basic pagination. Drivers, for example, offer 5 sort modes: **newest**, **oldest**, **popularity** (all-time points), **recent** (latest season points), and **wins** (total victories). Combined with nationality filtering and fuzzy name search, these enable complex discovery queries in a single GET request.

3.4 Referential Integrity Protection

DELETE operations enforce foreign key safety: attempting to delete a driver with associated race results returns **`409 Conflict`** with a descriptive message (e.g., `"Cannot delete driver: 150 race results are associated."`). The same protection applies to circuits and constructors.

3.5 Enriched Detail Views

Single-resource GET endpoints go beyond flat CRUD returns. Circuit detail includes the 5 most recent race winners; Constructor detail aggregates a full career summary (total entries, points, wins, podiums, active seasons) plus the top 5 highest-scoring drivers for that team — all computed at query time from normalised relational data.

3.6 Comprehensive Schema Design

The OpenAPI specification defines 40+ Pydantic schemas, each with **field-level descriptions**, **concrete examples**, and **type annotations**. This ensures the auto-generated Swagger UI is immediately usable for testing without consulting external documentation.

3.7 Role-Based Access Control

The API implements a two-tier permission model: **authenticated users** (JWT required) can create and update resources via POST/PUT, while **DELETE operations require admin privileges**. This mirrors real-world API governance patterns where destructive operations demand elevated permissions.

4. Data Source and Coverage

All data originates from the **Ergast F1 Dataset** (Kaggle mirror), imported via a CSV pipeline into a normalised PostgreSQL schema with 10 relational tables.

1950–2024	26,000+	850+	70+
Seasons Covered	Race Results	Unique Drivers	Circuits

5. Testing and Quality Assurance

The project includes a comprehensive pytest test suite covering all endpoint groups:

66	77%	7
Passing Tests	Code Coverage	Endpoint Groups

Test Coverage Includes

- Authentication flow (register, login, token validation, duplicate handling)
- Full CRUD lifecycle for Drivers, Circuits, and Constructors
- Pagination, search, and sort parameter validation
- Analytics endpoint correctness and edge cases (missing data, invalid IDs)
- Advanced analytics model output structure and boundary conditions
- Role-based access control enforcement (401/403 responses)

6. Quick Start

The live API is deployed at `https://f1-insight-api.onrender.com`. Interactive documentation is available at the `/docs` endpoint (Swagger UI) and `/redoc` (ReDoc).

To access protected endpoints: (1) Register via POST /api/v1/auth/register, (2) Login via POST /api/v1/auth/login to obtain a JWT, (3) Include the token as `Authorization: Bearer <token>` in subsequent requests.

F1 Insight API 1.0.0 OAS 3.1

/openapi.json

F1 Insight API

A comprehensive Formula 1 historical data analysis API covering **every season from 1950 to 2024**.

Features

- **CRUD Operations** – Manage drivers, constructors, and circuits with full pagination, filtering, and sorting
- **Career Analytics** – Lifetime stats, season progression curves, head-to-head driver comparisons
- **Strategy Analysis** – Pit stop strategies, circuit history, performance trends
- **Advanced Models** – Win probability prediction, multi-dimensional performance ratings, teammate battles, leaderboards

Authentication

Protected endpoints require a JWT Bearer token. Register via `POST /api/v1/auth/register`, then login via `POST /api/v1/auth/login` to receive your token.

Data Source

Ergast F1 Dataset – 10 tables, 26,000+ race results, 850+ drivers.

Authorize

Root Health check and API information.

GET / Root

GET /health Health Check

Authentication User registration and JWT-based login. Register an account and login to receive a Bearer token for protected endpoints.

POST /api/v1/auth/register Register a new user

Register a new user account. Username and email must be unique.

Parameters Try it out

No parameters

Request body required application/json

Example Value Schema

```
{
  "email": "lewis@example.com",
  "password": "securePass123",
  "username": "lewis_h"
}
```

Responses

Code	Description	Links
201	Successful Response Media type application/json Controls Accept header. **Example Value** Schema	*No links*

Code	Description	Links

```
{
  "id": 1,
  "username": "lewis_h",
  "email": "lewis@example.com",
  "role": "user"
}
```

400 Username or email already registered *No links*

Media type

application/json

Example Value

```
{
  "detail": "Username already registered"
}
```

422 Validation Error *No links*

Media type

application/json

Example Value Schema

```
{
  "detail": [
    {
      "loc": [
        "string",
        0
      ],
      "msg": "string",
      "type": "string"
    }
  ]
}
```

POST **/api/v1/auth/login** Login

Login with username and password to receive a JWT access token. Use the token as **Bearer <token>** in the Authorization header.

Parameters Try it out

No parameters

Request body required application/x-www-form-urlencoded

grant_type

string | (string | null)
pattern: password

username * required

string

password * required

string

scope

string

client_id

string | (string | null)

client_secret

string | (string | null)

Responses

Code	Description	Links
200	Successful Response Media type application/json Controls `Accept` header. Example Value Schema `{ "access_token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9...", "token_type": "bearer" }`	*No links*
401	Invalid credentials Media type application/json	*No links*

Code	Description	Links

	Example Value
	
	{
	"detail": "Incorrect username or password"
	}	

422	Validation Error
	
	Media type
	application/json
	
	Example Value Schema
	
	{
	"detail": [
	{
	"loc": [
	"string",
	0
	],
	"msg": "string",
	"type": "string"
	}
	]
	}
		No links

Drivers

CRUD operations for F1 drivers. Supports pagination, nationality filter, name search, and sorting by newest/oldest/popularity/recent/wins. POST/PUT require authentication; DELETE requires admin.

GET **/api/v1/drivers/** List drivers

Get a paginated list of drivers with optional filters and smart sorting.

Parameters

Try it out

Name	Description
page **integer** ***(query)***	Page number *Default value* : 1 1 ***minimum: 1***

Name	Description
per_page **integer** ***(query)***	Items per page *Default value* : 20 20 ***maximum: 100*** ***minimum: 1***
nationality **string | (string | null)** ***(query)***	Filter by nationality nationality
search **string | (string | null)** ***(query)***	Search by name, surname, or driver_ref search
sort_by **string | (string | null)** ***(query)***	Sort by: 'newest', 'oldest', 'popularity' (all-time points), 'recent' (latest season points), 'wins' (total wins) *Default value* : newest newest

Responses

Code	Description	Links
200	Successful Response Media type application/json Controls `Accept` header. **Example Value** Schema	*No links*


```
[
  {
    "driver_ref": "hamilton",
    "number": 44,
    "code": "HAM",
    "forename": "Lewis",
    "surname": "Hamilton",
```

Code	Description	Links


```
    "dob": "1985-01-07",
    "nationality": "British",
    "url": "http://en.wikipedia.org/wiki/Lewis_Hamilton",
    "driver_id": 1
  }
]
```


422	Validation Error	*No links*

Media type

application/json

Example Value Schema


```
{
  "detail": [
    {
      "loc": [
        "string",
        0
      ],
      "msg": "string",
      "type": "string"
    }
  ]
}
```


POST **/api/v1/drivers/** Create a new driver

Create a new driver (requires authentication).

Parameters Try it out

No parameters

Request body required application/json

Example Value Schema


```
{
  "code": "HAM",
  "dob": "1985-01-07",
  "driver_ref": "hamilton",
  "forename": "Lewis",
  "nationality": "British",
  "number": 44,
  "surname": "Hamilton",
```

```
  "url": "http://en.wikipedia.org/wiki/Lewis_Hamilton"
}
```


Responses

Code	Description	Links
201	Successful Response	*No links*

Media type

application/json

Controls `Accept` header.

Example Value Schema


```
{
  "driver_ref": "hamilton",
  "number": 44,
  "code": "HAM",
  "forename": "Lewis",
  "surname": "Hamilton",
  "dob": "1985-01-07",
  "nationality": "British",
  "url": "http://en.wikipedia.org/wiki/Lewis_Hamilton",
  "driver_id": 1
}
```


401	Authentication required	*No links*
409	Driver ref already exists	*No links*

Media type

application/json

Example Value


```
{
  "detail": "Driver with ref 'hamilton' already exists"
}
```


422	Validation Error	*No links*

Media type

application/json

Example Value Schema


```
{
  "detail": [
    {
      "loc": [
```

Code	Description	Links


```
            "string",
            0
          ],
          "msg": "string",
          "type": "string"
        }
      ]
    }
```


GET `/api/v1/drivers/{driver_id}` Get driver by ID

Get a single driver by ID.

Parameters

Try it out

Name	Description
driver_id * required integer *(path)*	driver_id

Responses

Code	Description	Links
200	Successful Response Media type application/json Controls Accept header. Example Value Schema	*No links*


```
{
  "driver_ref": "hamilton",
  "number": 44,
  "code": "HAM",
  "forename": "Lewis",
  "surname": "Hamilton",
  "dob": "1985-01-07",
  "nationality": "British",
  "url": "http://en.wikipedia.org/wiki/Lewis_Hamilton",
```

Code	Description	Links

```
  "driver_id": 1
}
```

404	Driver not found	*No links*

Media type

application/json

Example Value

```
{
  "detail": "Driver not found"
}
```

422	Validation Error	*No links*

Media type

application/json

Example Value Schema

```
{
  "detail": [
    {
      "loc": [
        "string",
        0
      ],
      "msg": "string",
      "type": "string"
    }
  ]
}
```

PUT **/api/v1/drivers/{driver_id}** Update a driver

Update a driver (requires authentication).

Parameters

Try it out

Name	Description
driver_id * required **integer** ***(path)***	driver_id

Request body required

application/json

Example Value Schema

```
{
  "nationality": "British",
  "number": 44
}
```

Responses

Code	Description	Links

200	Successful Response
	
	Media type
	application/json
	Controls Accept header.
	
	Example Value Schema
	
	{
	"driver_ref": "hamilton",
	"number": 44,
	"code": "HAM",
	"forename": "Lewis",
	"surname": "Hamilton",
	"dob": "1985-01-07",
	"nationality": "British",
	"url": "http://en.wikipedia.org/wiki/Lewis_Hamilton",
	"driver_id": 1
	}
		No links

400	No fields to update
	
	Media type
	application/json
	
	Example Value
	
	{
	"detail": "No fields to update"
	}
		No links

401	Authentication required	*No links*
404	Driver not found	*No links*
422	Validation Error	*No links*

Code	Description	Links

Media type

application/json

Example Value Schema


```
{
  "detail": [
    {
      "loc": [
        "string",
        0
      ],
      "msg": "string",
      "type": "string"
    }
  ]
}
```


DELETE **/api/v1/drivers/{driver_id}** Delete a driver

Delete a driver (requires admin privileges).

Parameters

Try it out

Name	Description
driver_id * required **integer** ***(path)***	driver_id

Responses

Code	Description	Links
204	Successful Response	*No links*
401	Authentication required	*No links*
403	Admin privileges required	*No links*

Code	Description	Links
404	Driver not found	*No links*
409	Cannot delete: associated race results exist	*No links*

Media type

application/json

Example Value

```
{
  "detail": "Cannot delete driver: 150 race results are assoc
iated. Remove associated results first."
}
```

422	Validation Error	*No links*

Media type

application/json

Example Value Schema

```
{
  "detail": [
    {
      "loc": [
        "string",
        0
      ],
      "msg": "string",
      "type": "string"
    }
  ]
}
```

Circuits

CRUD operations for F1 circuits. List includes total races per circuit. Detail view shows the 5 most recent race winners. POST/PUT require authentication; DELETE requires admin.

GET **/api/v1/circuits/** List circuits with stats

Get a paginated list of circuits with optional filters and sorting.

Parameters

Try it out

Name	Description
page **integer** ***(query)***	Page number *Default value* : 1 1 ***minimum: 1***
per_page **integer** ***(query)***	Items per page *Default value* : 20 20 ***maximum: 100*** ***minimum: 1***
country **string | (string | null)** ***(query)***	Filter by country country
search **string | (string | null)** ***(query)***	Search by circuit name or location search
sort_by **string | (string | null)** ***(query)***	Sort by: 'name' (alphabetical), 'country', 'most_races' (total races held), 'recent' (most recently used) *Default value* : name name

Responses

Code	Description	Links
200	Successful Response Media type application/json Controls `Accept` header.	*No links*

Code	Description	Links

Example Value Schema


```
[
  {
    "circuit_ref": "silverstone",
    "name": "Silverstone Circuit",
    "location": "Silverstone",
    "country": "UK",
    "lat": 52.0786,
    "lng": -1.01694,
    "alt": 153,
    "url": "http://en.wikipedia.org/wiki/Silverstone_Circui
t",
    "circuit_id": 9,
    "total_races": 59
  }
]
```


422 Validation Error *No links*

Media type

application/json

Example Value Schema


```
{
  "detail": [
    {
      "loc": [
        "string",
        0
      ],
      "msg": "string",
      "type": "string"
    }
  ]
}
```


POST **/api/v1/circuits/** Create a new circuit

Create a new circuit (requires authentication).

Parameters Try it out

No parameters

Request body required application/json

Example Value Schema


```
{
  "alt": 153,
  "circuit_ref": "silverstone",
  "country": "UK",
  "lat": 52.0786,
  "lng": -1.01694,
  "location": "Silverstone",
  "name": "Silverstone Circuit",
  "url": "http://en.wikipedia.org/wiki/Silverstone_Circuit"
}
```


Responses

Code	Description	Links
201	Successful Response	*No links*

Media type

application/json

Controls `Accept` header.

Example Value Schema


```
{
  "circuit_ref": "silverstone",
  "name": "Silverstone Circuit",
  "location": "Silverstone",
  "country": "UK",
  "lat": 52.0786,
  "lng": -1.01694,
  "alt": 153,
  "url": "http://en.wikipedia.org/wiki/Silverstone_Circuit",
  "circuit_id": 9
}
```


401	Authentication required	*No links*
409	Circuit ref already exists	*No links*

Media type

application/json

Example Value


```
{
  "detail": "Circuit with ref 'silverstone' already exists"
}
```

Code	Description	Links
422	Validation Error Media type application/json **Example Value** Schema	*No links*

```
{
  "detail": [
    {
      "loc": [
        "string",
        0
      ],
      "msg": "string",
      "type": "string"
    }
  ]
}
```

GET **/api/v1/circuits/{circuit_id}** Get circuit detail with race history

Get a single circuit by ID with recent race history.

Returns circuit details plus the 5 most recent races held at this circuit, including the winner of each race.

Parameters

Try it out

Name	Description
circuit_id * required **integer** ***(path)***	circuit_id

Responses

Code	Description	Links
200	Successful Response Media type	*No links*

Code	Description	Links

application/json

Controls `Accept` header.

Example Value Schema


```
{
  "circuit_id": 9,
  "circuit_ref": "silverstone",
  "name": "Silverstone Circuit",
  "location": "Silverstone",
  "country": "UK",
  "lat": 52.0786,
  "lng": -1.01694,
  "alt": 153,
  "url": "string",
  "total_races_held": 59,
  "recent_races": [
    {
      "race_id": 1100,
      "year": 2024,
      "round": 12,
      "race_name": "British Grand Prix",
      "date": "2024-07-07",
      "winner": "Lewis Hamilton"
    }
  ]
}
```


404	Circuit not found	*No links*

Media type

application/json

Example Value


```
{
  "detail": "Circuit not found"
}
```


422	Validation Error	*No links*

Media type

application/json

Example Value Schema


```
{
  "detail": [
    {
      "loc": [
        "string",
        0
      ],
      "msg": "string",
      "type": "string"
    }
```

Code	Description	Links
	] }	

PUT `/api/v1/circuits/{circuit_id}` Update a circuit

Update a circuit (requires authentication).

Parameters Try it out

Name	Description
circuit_id * required *integer* *(path)*	circuit_id

Request body required application/json

Example Value Schema

```
{
  "country": "UK",
  "name": "Silverstone Circuit"
}
```

Responses

Code	Description	Links
200	Successful Response Media type application/json Controls `Accept` header. Example Value Schema { "circuit_ref": "silverstone", "name": "Silverstone Circuit",	*No links*

Code	Description	Links


```
  "location": "Silverstone",
  "country": "UK",
  "lat": 52.0786,
  "lng": -1.01694,
  "alt": 153,
  "url": "http://en.wikipedia.org/wiki/Silverstone_Circuit",
  "circuit_id": 9
}
```


400	No fields to update	*No links*
401	Authentication required	*No links*
404	Circuit not found	*No links*
422	Validation Error	*No links*

Media type

application/json

Example Value Schema


```
{
  "detail": [
    {
      "loc": [
        "string",
        0
      ],
      "msg": "string",
      "type": "string"
    }
  ]
}
```


DELETE **/api/v1/circuits/{circuit_id}** Delete a circuit

Delete a circuit (requires admin privileges).

Parameters Try it out

Name	Description
circuit_id * required **integer** ***(path)***	circuit_id

Responses

Code	Description	Links
204	Successful Response	*No links*
401	Authentication required	*No links*
403	Admin privileges required	*No links*
404	Circuit not found	*No links*
409	Cannot delete: associated races exist	*No links*

Media type

application/json

Example Value

```
{
  "detail": "Cannot delete circuit: 59 races are associated w
ith it. Remove associated races first."
}
```

422	Validation Error	*No links*

Media type

application/json

Example Value Schema

```
{
  "detail": [
    {
      "loc": [
        "string",
        0
      ],
      "msg": "string",
      "type": "string"
    }
  ]
}
```

Constructors

CRUD operations for F1 constructors (teams). List includes total points and wins. Detail view shows career summary and top 5 drivers. POST/PUT require authentication; DELETE requires admin.

GET `/api/v1/constructors/` List constructors with stats

Get a paginated list of constructors with optional filters and sorting.

Parameters Try it out

Name	Description
page **integer** ***(query)***	Page number *Default value* : 1 1 ***minimum: 1***
per_page **integer** ***(query)***	Items per page *Default value* : 20 20 ***maximum: 100*** ***minimum: 1***
nationality **string | (string | null)** ***(query)***	Filter by nationality nationality
search **string | (string | null)** ***(query)***	Search by constructor name or ref search
sort_by **string | (string | null)** ***(query)***	Sort by: 'name' (alphabetical), 'nationality', 'total_points' (all-time points), 'wins' (total wins), 'recent' (most recently active) *Default value* : name

Name	Description
	name

Responses

Code	Description	Links
200	Successful Response Media type application/json Controls `Accept` header. Example Value Schema	*No links*
422	Validation Error Media type application/json Example Value Schema	*No links*


```
[
  {
    "constructor_ref": "mercedes",
    "name": "Mercedes",
    "nationality": "German",
    "url": "http://en.wikipedia.org/wiki/Mercedes-Benz_in_Formula_One",
    "constructor_id": 1,
    "total_points": 7005.5,
    "wins": 125
  }
]
```

```
{
  "detail": [
    {
      "loc": [
        "string",
        0
      ],
      "msg": "string",
      "type": "string"
    }
  ]
}
```

POST `/api/v1/constructors/` Create a new constructor

Create a new constructor (requires authentication).

Parameters Try it out

No parameters

Request body required application/json

Example Value Schema

```
{
  "constructor_ref": "mercedes",
  "name": "Mercedes",
  "nationality": "German",
  "url": "http://en.wikipedia.org/wiki/Mercedes-Benz_in_Formula_One"
}
```

Responses

Code	Description	Links
201	Successful Response Media type application/json Controls `Accept` header. Example Value Schema `{ "constructor_ref": "mercedes", "name": "Mercedes", "nationality": "German", "url": "http://en.wikipedia.org/wiki/Mercedes-Benz_in_Formula_One", "constructor_id": 1 }`	*No links*
401	Authentication required	*No links*
409	Constructor ref already exists Media type	*No links*

Code	Description	Links

application/json

Example Value

```
{
  "detail": "Constructor with ref 'mercedes' already exists"
}
```

422	Validation Error	*No links*

Media type

application/json

Example Value Schema

```
{
  "detail": [
    {
      "loc": [
        "string",
        0
      ],
      "msg": "string",
      "type": "string"
    }
  ]
}
```

GET **/api/v1/constructors/{constructor_id}**

Get constructor detail with career summary

Get a single constructor by ID with career summary and notable drivers.

Returns constructor details plus all-time stats (total races, wins, points) and the top 5 highest-scoring drivers who raced for this team.

Parameters

Try it out

Name	Description
constructor_id * required **integer** ***(path)***	constructor_id

Responses

Code	Description	Links
200	Successful Response	*No links*

Media type

application/json

Controls `Accept` header.

Example Value Schema


```
{
  "constructor_id": 1,
  "constructor_ref": "mercedes",
  "name": "Mercedes",
  "nationality": "German",
  "url": "string",
  "career_summary": {
    "total_entries": 268,
    "total_points": 7005.5,
    "wins": 125,
    "podiums": 275,
    "first_season": 2010,
    "last_season": 2024
  },
  "top_drivers": [
    {
      "driver": "Lewis Hamilton",
      "races_for_team": 192,
      "points_for_team": 2560.5,
      "wins_for_team": 84
    }
  ]
}
```


404	Constructor not found	*No links*

Media type

application/json

Example Value


```
{
  "detail": "Constructor not found"
}
```


422	Validation Error	*No links*

Media type

application/json

Example Value Schema


```
{
  "detail": [
```

Code	Description	Links

```
      {
        "loc": [
          "string",
          0
        ],
        "msg": "string",
        "type": "string"
      }
    ]
  }
```

PUT `/api/v1/constructors/{constructor_id}` Update a constructor

Update a constructor (requires authentication).

Parameters Try it out

Name	Description
constructor_id * required integer *(path)*	constructor_id

Request body required application/json

Example Value Schema

```
{
  "name": "Mercedes AMG Petronas",
  "nationality": "German"
}
```

Responses

Code	Description	Links
200	Successful Response Media type application/json	*No links*

Code	Description	Links
	Controls `Accept` header. **Example Value** Schema `{ "constructor_ref": "mercedes", "name": "Mercedes", "nationality": "German", "url": "http://en.wikipedia.org/wiki/Mercedes-Benz_in_Formula_One", "constructor_id": 1 }`	
400	No fields to update	*No links*
401	Authentication required	*No links*
404	Constructor not found	*No links*
422	Validation Error Media type application/json **Example Value** Schema `{ "detail": [ { "loc": [ "string", 0 ], "msg": "string", "type": "string" } ] }`	*No links*

DELETE `/api/v1/constructors/{constructor_id}` Delete a constructor

Delete a constructor (requires admin privileges).

Parameters Try it out

Name	Description
constructor_id * required **integer** ***(path)***	constructor_id

Responses

Code	Description	Links
204	Successful Response	*No links*
401	Authentication required	*No links*
403	Admin privileges required	*No links*
404	Constructor not found	*No links*
409	Cannot delete: associated results exist	*No links*

Media type

application/json

Example Value

```
{
  "detail": "Cannot delete constructor: 268 race results are
associated with it. Remove associated results first."
}
```

422	Validation Error	*No links*

Media type

application/json

Example Value Schema

```
{
  "detail": [
    {
      "loc": [
        "string",
        0
      ],
      "msg": "string",
      "type": "string"
    }
```

Code	Description	Links

```
  ]
}
```

Analytics

Core F1 data analysis: career stats, season progression curves, head-to-head driver comparisons, pit stop strategy analysis, and circuit historical records. All computed from raw race data (no pre-aggregated tables).

GET `/api/v1/analytics/drivers/{driver_id}/career-stats` Get driver career stats

Get comprehensive career statistics for a driver.

Includes total races, wins, podiums, poles, points, DNFs, win/podium rates, teams driven for, and active seasons.

Parameters

Try it out

Name	Description
driver_id * required **integer** ***(path)***	driver_id

Responses

Code	Description	Links
200	Successful Response Media type application/json Controls `Accept` header. Example Value Schema	*No links*

```
  driver : {
    "id": 1,
    "name": "Lewis Hamilton",
```

Code | Links


```
  },
  "career": {
    "total_races": 340,
    "wins": 103,
    "podiums": 197,
    "pole_positions": 104,
    "total_points": 4639.5,
    "dnfs": 30,
    "best_finish": 1,
    "win_rate": 30.29,
    "podium_rate": 57.94
  },
  "teams": [
    "McLaren",
    "Mercedes",
    "Ferrari"
  ],
  "seasons": [
    0
  ]
}
```


404 Driver not found or no race data — *No links*

Media type

application/json

Example Value


```
{
  "detail": "Driver not found"
}
```


422 Validation Error — *No links*

Media type

application/json

Example Value Schema


```
{
  "detail": [
    {
      "loc": [
        "string",
        0
      ],
      "msg": "string",
      "type": "string"
    }
  ]
}
```


GET **/api/v1/analytics/drivers/{driver_id}/season-progression**

Get season points progression

Get a driver's points accumulation across a season. Returns round-by-round cumulative points for charting progression curves.

Parameters

Try it out

Name	Description
driver_id * required integer (path)	driver_id
season * required integer (query)	Season year season

Responses

Code	Description	Links
200	Successful Response Media type application/json Controls Accept header. Example Value Schema	No links
404	Driver not found or no data for season Media type	No links


```
{
  "driver": "Lewis Hamilton",
  "season": 2020,
  "total_points": 347,
  "rounds": [
    {
      "round": 1,
      "race_name": "Bahrain Grand Prix",
      "position": 1,
      "points_scored": 25,
      "cumulative_points": 25
    }
  ]
}
```

Code	Description	Links

application/json

Example Value

```
{
  "detail": "No data for this driver in the specified season"
}
```

422	Validation Error	*No links*

Media type

application/json

Example Value Schema

```
{
  "detail": [
    {
      "loc": [
        "string",
        0
      ],
      "msg": "string",
      "type": "string"
    }
  ]
}
```

GET **/api/v1/analytics/drivers/compare** Compare two drivers head-to-head

Head-to-head comparison between two drivers. Compares wins, points, podiums, and direct race encounters.

Parameters

Try it out

Name	Description
d1 * required **integer** ***(query)***	First driver ID d1
d2 * required **integer** ***(query)***	Second driver ID d2
season	Optional season filter

Name	Description
integer | (integer | null) *(query)*	season

Responses

Code	Description	Links
200	Successful Response	*No links*

Media type

application/json

Controls `Accept` header.

Example Value Schema


```
{
  "driver_1": {
    "name": "Lewis Hamilton",
    "stats": {
      "total_races": 340,
      "wins": 103,
      "podiums": 197,
      "total_points": 4639.5,
      "dnfs": 30,
      "avg_finish": 4.52
    }
  },
  "driver_2": {
    "name": "Lewis Hamilton",
    "stats": {
      "total_races": 340,
      "wins": 103,
      "podiums": 197,
      "total_points": 4639.5,
      "dnfs": 30,
      "avg_finish": 4.52
    }
  },
  "head_to_head": {
    "common_races": 200
```


404	One or both drivers not found	*No links*

Media type

application/json

Example Value


```
{
  "detail": "One or both drivers not found"
```

Code	Description	Links
	}	
422	Validation Error Media type application/json Example Value Schema	*No links*

```
{
  "detail": [
    {
      "loc": [
        "string",
        0
      ],
      "msg": "string",
      "type": "string"
    }
  ]
}
```

GET **/api/v1/analytics/races/{race_id}/pit-stop-analysis**
Analyse pit stop strategies

Analyse pit stop strategies and their impact on race results for a given race.

Shows per-driver stop count, total pit time, and strategy summary with average finishing position by number of stops.

Parameters Try it out

Name	Description
race_id * required **integer** ***(path)***	race_id

Responses

Code	Description	Links
200	Successful Response	*No links*

Code	Description	Links

Media type

application/json

Controls `Accept` header.

Example Value Schema


```
{
  "race": "British Grand Prix",
  "season": 2024,
  "strategy_summary": {
    "additionalProp1": {
      "drivers": 12,
      "avg_finish_position": 6.5
    },
    "additionalProp2": {
      "drivers": 12,
      "avg_finish_position": 6.5
    },
    "additionalProp3": {
      "drivers": 12,
      "avg_finish_position": 6.5
    }
  },
  "driver_details": [
    {
      "driver": "Lewis Hamilton",
      "grid_position": 1,
      "final_position": 1,
      "num_stops": 2,
      "total_pit_time_ms": 46280,
      "stops_detail": [
```


404	Race not found or no pit stop data	*No links*

Media type

application/json

Example Value


```
{
  "detail": "No pit stop data available for this race"
}
```


422	Validation Error	*No links*

Media type

application/json

Example Value Schema


```
{
  "detail": [
    {
      "loc": [
        "string",
        0
```

Code	Description	Links


```
        ],
        "msg": "string",
        "type": "string"
      }
    ]
  }
```


GET `/api/v1/analytics/circuits/{circuit_id}/history` Get circuit historical stats

Get historical statistics for a circuit.

Includes total races held, first/last race year, and top 5 most successful drivers by wins at this circuit.

Parameters

Try it out

Name	Description
circuit_id * required integer *(path)*	circuit_id

Responses

Code	Description	Links
200	Successful Response Media type application/json Controls `Accept` header. Example Value Schema	*No links*


```
{
  "circuit": {
    "name": "Silverstone Circuit",
    "location": "Silverstone",
    "country": "UK"
  },
  "statistics": {
    "total_races": 59,
    "first_race_year": 1950,
```

Code	Description	Links


```
    "last_race_year": 2024
  },
  "most_successful_drivers": [
    {
      "driver": "Lewis Hamilton",
      "wins": 8
    }
  ]
}
```


404 No races found for this circuit *No links*

Media type

application/json

Example Value


```
{
  "detail": "No races found for this circuit"
}
```


422 Validation Error *No links*

Media type

application/json

Example Value Schema


```
{
  "detail": [
    {
      "loc": [
        "string",
        0
      ],
      "msg": "string",
      "type": "string"
    }
  ]
}
```


Advanced Analytics

Advanced analytical models: win probability prediction (4-factor weighted model), multi-dimensional performance ratings (0-100), teammate head-to-head battle comparisons, and dynamic leaderboards by various metrics.

GET **/api/v1/analytics/win-probability** Predict win probability

Predict a driver's win probability at a specific circuit based on historical data.

Combines multiple factors:

- Historical win rate at this circuit
- Recent form (last 5 races overall)
- Circuit-specific finishing positions
- Grid-to-finish conversion rate

Returns a probability score (0–100%) with breakdown of contributing factors.

Parameters

Try it out

Name	Description
driver_id * required **integer** ***(query)***	Driver ID driver_id
circuit_id * required **integer** ***(query)***	Circuit ID circuit_id

Responses

Code	Description	Links
200	Successful Response Media type application/json Controls `Accept` header.	*No links*

Example Value Schema


```
{
  "driver": "Lewis Hamilton",
  "circuit": "Silverstone Circuit",
  "win_probability": "32.5%",
  "factors": {
    "circuit_history": {
      "weight": "35%",
      "score": 53.3,
      "detail": {
        "races_at_circuit": 15,
        "wins_at_circuit": 8,
        "podiums_at_circuit": 12,
        "avg_finish_at_circuit": 2.3
      }
    },
```

Code	Description	Links

```
    "recent_form": {
      "weight": "30%",
      "score": 72.1,
      "detail": {
        "last_5_races_avg_finish": 3.2,
        "last_5_wins": 2,
        "last_5_podiums": 4
      }
    }.
```

404 Driver or circuit not found *No links*

Media type

application/json

Example Value

```
{
  "detail": "Driver not found"
}
```

422 Validation Error *No links*

Media type

application/json

Example Value Schema

```
{
  "detail": [
    {
      "loc": [
        "string",
        0
      ],
      "msg": "string",
      "type": "string"
    }
  ]
}
```

GET **/api/v1/analytics/drivers/{driver_id}/performance-summary**

Get multi-dimensional performance rating

Generate a comprehensive multi-dimensional performance summary for a driver.

Analyses qualifying pace, race pace, consistency, overtaking ability, and pit stop efficiency to produce an overall rating out of 100.

Parameters Try it out

Name	Description
driver_id * required integer *(path)*	driver_id
season integer | (integer | null) *(query)*	Filter by season (default: all-time) season

Responses

Code	Description	Links
200	Successful Response Media type application/json Controls Accept header. Example Value Schema	*No links*
404	Driver not found or no data Media type	*No links*


```
{
  "driver": "Lewis Hamilton",
  "season": 2020,
  "overall_rating": 82.5,
  "total_races": 17,
  "dimensions": {
    "qualifying_pace": {
      "score": 85.3,
      "weight": "25%",
      "detail": {
        "avg_qualifying_position": 3.2,
        "pole_positions": 104,
        "top_3_qualifying": 250
      }
    },
    "race_pace": {
      "score": 81.6,
      "weight": "30%",
      "detail": {
        "avg_finish_position": 4.5,
        "wins": 103,
        "podiums": 197,
        "points_per_race": 13.6
      }
    }
```

Code	Description	Links

	application/json
	
	Example Value
	
	{
	"detail": "No data found for this driver"
	}	

422	Validation Error
	
	Media type
	application/json
	
	Example Value Schema
	
	{
	"detail": [
	{
	"loc": [
	"string",
	0
	],
	"msg": "string",
	"type": "string"
	}
	]
	}
		No links

GET **/api/v1/analytics/teammate-battle** Compare driver vs teammate(s)

Head-to-head comparison between a driver and their teammate(s) in a specific season.

Compares qualifying positions, race results, and points scored in races where both drivers participated.

Parameters Try it out

Name	Description
driver_id * required **integer** ***(query)***	Driver ID driver_id

Name	Description
season * required **integer** ***(query)***	Season year season

Responses

Code	Description	Links
200	Successful Response Media type application/json Controls `Accept` header.	*No links*

Example Value Schema


```
{
  "driver": "Lewis Hamilton",
  "season": 2020,
  "battles": [
    {
      "teammate": "Valtteri Bottas",
      "common_races": 20,
      "qualifying": {
        "driver_wins": 12,
        "teammate_wins": 8,
        "winner": "Hamilton"
      },
      "race": {
        "driver_wins": 12,
        "teammate_wins": 8,
        "winner": "Hamilton"
      },
      "points": {
        "driver_points": 347,
        "teammate_points": 230.5,
        "winner": "Hamilton"
      }
    }
  ]
}
```


404	Driver not found or no data for season Media type application/json	*No links*

Example Value

Code	Description	Links

```
{
  "detail": "No data for this driver in 2024"
}
```

422 Validation Error *No links*

Media type

application/json

Example Value Schema

```
{
  "detail": [
    {
      "loc": [
        "string",
        0
      ],
      "msg": "string",
      "type": "string"
    }
  ]
}
```

GET **/api/v1/analytics/leaderboard** Get driver leaderboard

Dynamic leaderboard that ranks drivers by various metrics.

Supports filtering by season and ranking by total points, wins, podiums, or win rate.

Parameters Try it out

Name	Description
season **integer | (integer | null)** ***(query)***	Season year (default: all-time) season
metric **string** ***(query)***	Ranking metric: 'points', 'wins', 'podiums', 'win_rate' *Default value* : points points
limit	Number of drivers to return

Name	Description
integer *(query)*	*Default value* : 20 20 ***maximum: 100*** ***minimum: 1***

Responses

Code	Description	Links
200	Successful Response Media type application/json Controls `Accept` header. **Example Value** Schema	*No links*
422	Invalid metric or parameter	*No links*


```
{
  "season": "all-time",
  "metric": "points",
  "total_drivers": 20,
  "leaderboard": [
    {
      "rank": 1,
      "driver": "Lewis Hamilton",
      "nationality": "British",
      "races": 340,
      "wins": 103,
      "podiums": 197,
      "total_points": 4639.5,
      "win_rate": "30.3%"
    }
  ]
}
```


Schemas

BattleScore › Expand all **object**

Body_login_api_v1_auth_login_post > Expand all **object**

CareerBaselineFactor > Expand all **object**

CareerBaselineFactorDetail > Expand all **object**

CareerStatistics ^ Collapse all **object**

Comprehensive career statistics.

- **total_races*** > Expand all **integer**
- **wins*** > Expand all **integer**
- **podiums*** > Expand all **integer**
- **pole_positions*** > Expand all **integer**
- **total_points*** > Expand all **number**
- **dnfs*** > Expand all **integer**
- **best_finish** > Expand all **(integer | null)**
- **win_rate*** > Expand all **number**
- **podium_rate*** > Expand all **number**

CareerStatsResponse ^ Collapse all **object**

Full career statistics for a driver.

- **driver*** > Expand all **object**
- **career*** > Expand all **object**
- **teams*** > Expand all **array<string>**
- **seasons*** > Expand all **array<integer>**

CircuitCreate ^ Collapse all **object**

- **circuit_ref*** > Expand all **string**
- **name*** > Expand all **string**
- **location** > Expand all **(string | null)**
- **country** > Expand all **(string | null)**
- **lat** > Expand all **(number | null)**
- **lng** > Expand all **(number | null)**
- **alt** > Expand all **(number | null)**
- **url** > Expand all **(string | null)**

Example > Expand all **object**

CircuitDetailResponse ^ Collapse all **object**

Circuit details with race history and aggregated stats.

circuit_id* > Expand all **integer**

circuit_ref* > Expand all **string**

name* > Expand all **string**

location > Expand all **(string | null)**

country > Expand all **(string | null)**

lat > Expand all **(number | null)**

lng > Expand all **(number | null)**

alt > Expand all **(number | null)**

url > Expand all **(string | null)**

total_races_held* > Expand all **integer**

recent_races* > Expand all **array<object>**

CircuitHistoryFactor > Expand all **object**

CircuitHistoryFactorDetail > Expand all **object**

CircuitHistoryInfo > Expand all **object**

CircuitHistoryResponse ^ Collapse all **object**

Historical statistics and most successful drivers at a circuit.

circuit* > Expand all **object**

statistics* > Expand all **object**

most_successful_drivers* > Expand all **array<object>**

CircuitHistoryStats > Expand all **object**

CircuitListResponse > Expand all **object**

CircuitRaceEntry > Expand all **object**

CircuitResponse ^ Collapse all **object**

- **circuit_ref*** > Expand all **string**
- **name*** > Expand all **string**
- **location** > Expand all **(string | null)**
- **country** > Expand all **(string | null)**
- **lat** > Expand all **(number | null)**
- **lng** > Expand all **(number | null)**
- **alt** > Expand all **(number | null)**
- **url** > Expand all **(string | null)**
- **circuit_id*** > Expand all **integer**

CircuitUpdate > Expand all **object**

CircuitWinnerEntry > Expand all **object**

ConsistencyDetail > Expand all **object**

ConsistencyDimension > Expand all **object**

ConstructorCareerSummary > Expand all **object**

ConstructorCreate ^ Collapse all **object**

- **constructor_ref*** > Expand all **string**
- **name*** > Expand all **string**
- **nationality** > Expand all **(string | null)**
- **url** > Expand all **(string | null)**
- *Example* > Expand all **object**

ConstructorDetailResponse ^ Collapse all **object**

Constructor details with career summary and notable drivers.

- **constructor_id*** > Expand all **integer**
- **constructor_ref*** > Expand all **string**
- **name*** > Expand all **string**
- **nationality** > Expand all **(string | null)**

url > Expand all **(string | null)**

career_summary* > Expand all **object**

top_drivers* > Expand all **array<object>**

ConstructorListResponse > Expand all **object**

ConstructorResponse ^ Collapse all **object**

constructor_ref* > Expand all **string**

name* > Expand all **string**

nationality > Expand all **(string | null)**

url > Expand all **(string | null)**

constructor_id* > Expand all **integer**

ConstructorTopDriver > Expand all **object**

ConstructorUpdate > Expand all **object**

DriverCareerInfo > Expand all **object**

DriverCompareEntry > Expand all **object**

DriverCompareResponse ^ Collapse all **object**

Head-to-head comparison between two drivers.

driver_1* > Expand all **object**

driver_2* > Expand all **object**

head_to_head* > Expand all **object**

season_filter > Expand all **(integer | null)**

DriverCompareStats > Expand all **object**

DriverCreate ^ Collapse all **object**

driver_ref* > Expand all **string**

number > Expand all **(integer | null)**

code > Expand all **(string | null)**

forename* > Expand all **string**

surname* > Expand all **string**

dob > Expand all **(string | null)**

nationality > Expand all **(string | null)**

url > Expand all **(string | null)**

Example > Expand all **object**

DriverPitStopAnalysis > Expand all **object**

DriverResponse ^ Collapse all **object**

driver_ref* > Expand all **string**

number > Expand all **(integer | null)**

code > Expand all **(string | null)**

forename* > Expand all **string**

surname* > Expand all **string**

dob > Expand all **(string | null)**

nationality > Expand all **(string | null)**

url > Expand all **(string | null)**

driver_id* > Expand all **integer**

DriverUpdate ^ Collapse all **object**

number > Expand all **(integer | null)**

code > Expand all **(string | null)**

forename > Expand all **(string | null)**

surname > Expand all **(string | null)**

nationality > Expand all **(string | null)**

url > Expand all **(string | null)**

Example > Expand all **object**

GridConversionFactor > Expand all **object**

GridConversionFactorDetail > Expand all **object**

HTTPValidationError > Expand all **object**

HeadToHeadResult > Expand all **object**

LeaderboardEntry > Expand all **object**

LeaderboardResponse ^ Collapse all **object**

Dynamic leaderboard ranking drivers by a chosen metric.

- **season*** > Expand all **(integer | string)**
- **metric*** > Expand all **string**
- **total_drivers*** > Expand all **integer**
- **leaderboard*** > Expand all **array<object>**

OvertakingDetail > Expand all **object**

OvertakingDimension > Expand all **object**

PerformanceDimensions > Expand all **object**

PerformanceSummaryResponse ^ Collapse all **object**

Multi-dimensional performance rating for a driver.

- **driver*** > Expand all **string**
- **season*** > Expand all **(integer | string)**
- **overall_rating*** > Expand all **number**
- **total_races*** > Expand all **integer**
- **dimensions*** > Expand all **object**

PitStopAnalysisResponse ^ Collapse all **object**

Pit stop strategy analysis for a specific race.

- **race*** > Expand all **string**
- **season*** > Expand all **integer**
- **strategy_summary*** > Expand all **object**
- **driver_details*** > Expand all **array<object>**

PitStopDetailEntry > Expand all **object**

PitStopEfficiencyDetail > Expand all **object**

PitStopEfficiencyDimension > Expand all **object**

PointsBattleScore > Expand all **object**

QualifyingPaceDetail > Expand all **object**

QualifyingPaceDimension > Expand all **object**

RacePaceDetail > Expand all **object**

RacePaceDimension > Expand all **object**

RecentFormFactor > Expand all **object**

RecentFormFactorDetail > Expand all **object**

RoundProgressEntry > Expand all **object**

SeasonProgressionResponse ^ Collapse all **object**

A driver's round-by-round points accumulation across a season.

- **driver*** > Expand all **string**
- **season*** > Expand all **integer**
- **total_points*** > Expand all **number**
- **rounds*** > Expand all **array<object>**

StrategyStats > Expand all **object**

TeammateBattleEntry > Expand all **object**

TeammateBattleResponse ^ Collapse all **object**

Head-to-head comparison between a driver and their teammate(s) in a season.

driver* > Expand all **string**

season* > Expand all **integer**

battles* > Expand all **array<object>**

Token ^ Collapse all **object**

access_token* > Expand all **string**

token_type > Expand all **string**

UserCreate ^ Collapse all **object**

username* > Expand all **string** [3, 50] characters

email* > Expand all **string** ≤ 255 characters

password* > Expand all **string** ≥ 6 characters

Example > Expand all **object**

UserResponse ^ Collapse all **object**

id* > Expand all **integer**

username* > Expand all **string**

email* > Expand all **string**

role* > Expand all **string**

ValidationError > Expand all **object**

WinProbabilityFactors > Expand all **object**

WinProbabilityResponse ^ Collapse all **object**

Win probability prediction with weighted factor breakdown.

driver* > Expand all **string**

circuit* > Expand all **string**

win_probability* > Expand all **string**

factors* > Expand all **object**

note* > Expand all **string**